Aim - To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory - Ganache is a personal Ethereum blockchain used for testing and development. It allows developers to deploy contracts, develop decentralized applications (DApps), and run tests in a safe and deterministic environment.

Ganache provides:

- Instant mining of blocks
- Pre-funded test accounts
- Transaction history view
- Graphical interface to monitor blockchain activities

It is part of the Truffle Suite and is commonly used with Remix and Metamask for smart contract deployment.

Steps to Connect Ganache with Metamask and Remix IDE:

1. Install Ganache:
    - Download and install Ganache from the Truffle Suite website.
    - Launch it to start a local blockchain network with test Ether accounts.
2. Connect Ganache Accounts with Metamask:
    - Open Metamask and switch to a Custom RPC Network.
    - Enter the Ganache RPC server URL (e.g., HTTP://127.0.0.1:7545).
    - Import private keys of Ganache accounts into Metamask.
3. Connect Remix IDE with Metamask:
    - Open Remix IDE in the browser.
      Choose Injected Web3 as the environment in the "Deploy & Run Transactions" tab.
    - Remix will automatically connect to the Metamask account (linked with Ganache).
4. Create a Simple Solidity Smart Contract:
    - Write a Solidity program related to your mini project (e.g., Student Record, Donation Tracker, etc.).

```
pragma solidity ^0.8.0;
    contract SimpleStorage {
      uint public data;
      function set(uint x) public {
        data = x;
      }
```

5. Compile and Deploy the Contract:
   Compile using Remix.
    - Deploy via the Metamask account connected to Ganache.
    - Approve the transaction in Metamask.

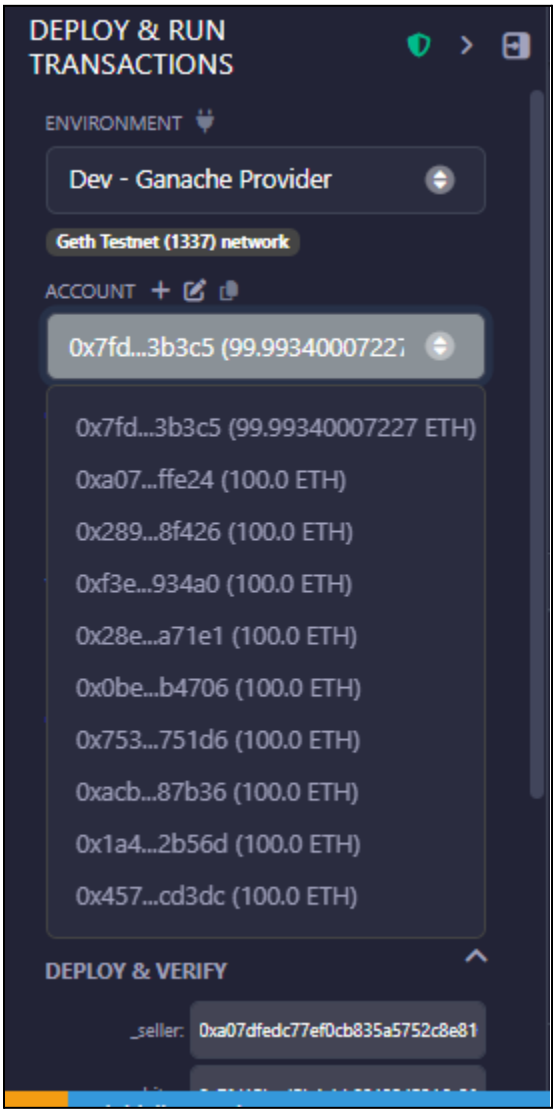
6. Check Transaction Details on Ganache:
    - View mined transactions, block details, and balances in the Ganache interface.
7. Interact with the Smart Contract:
    - Use Remix to call contract functions (set() or get()).
    - Observe the state changes and corresponding transactions on Ganache.

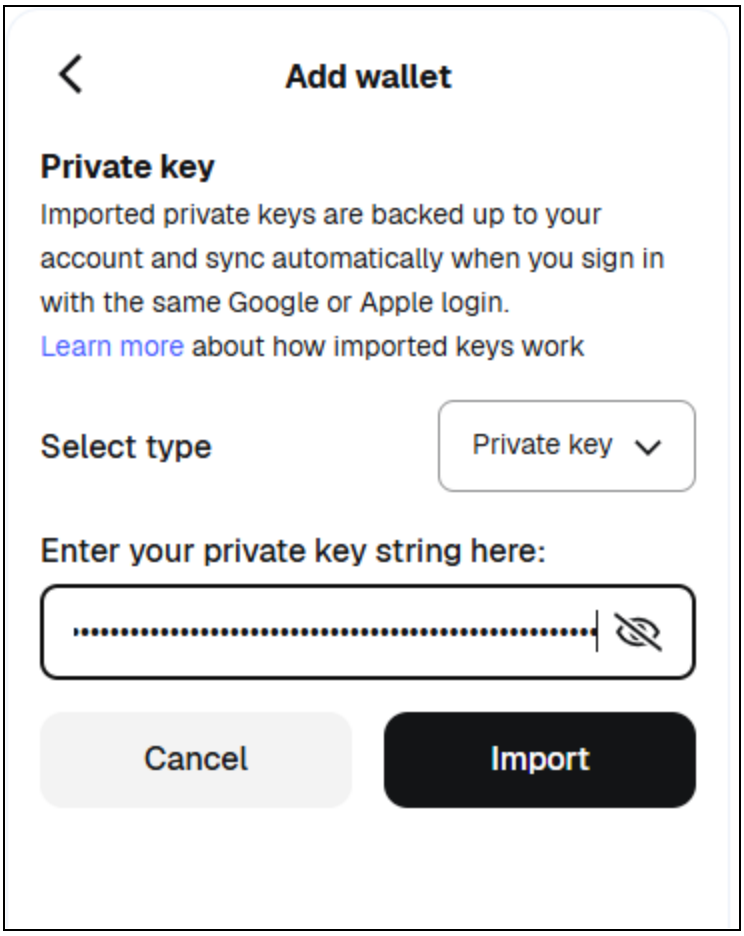
Code -

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract Escrow {

    address public buyer;
    address payable public seller;
    address public arbiter;

    bool public buyerApproved;
    bool public sellerApproved;

    constructor(address payable _seller, address _arbiter) payable {
        buyer = msg.sender;
        seller = _seller;
        arbiter = _arbiter;
    }

    function approve() public {
        require(msg.sender == buyer || msg.sender == seller, "Not authorized");

        if (msg.sender == buyer) {
            buyerApproved = true;
        }

        if (msg.sender == seller) {
            sellerApproved = true;
        }

        if (buyerApproved && sellerApproved) {

            uint amount = address(this).balance;

            (bool success, ) = seller.call{value: amount}("");

            require(success, "Transfer failed");
        }
    }
```

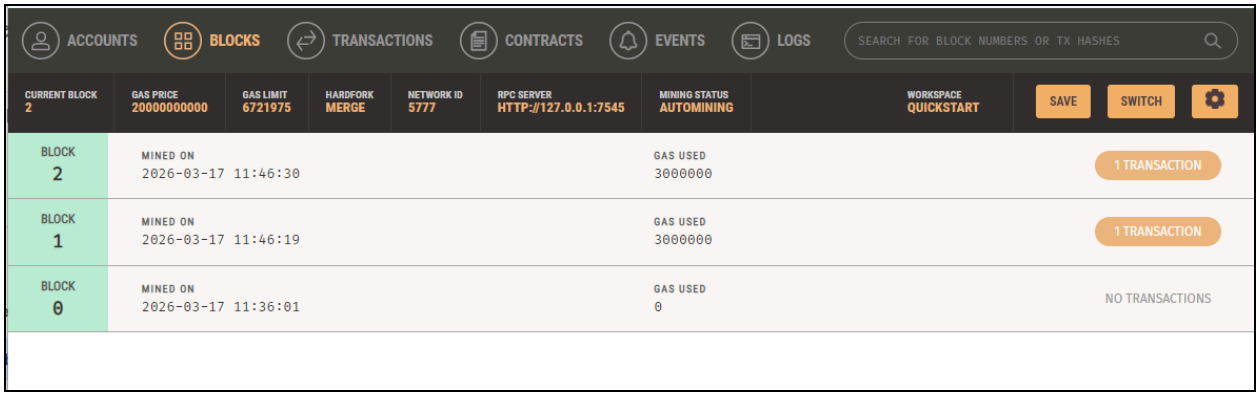
}

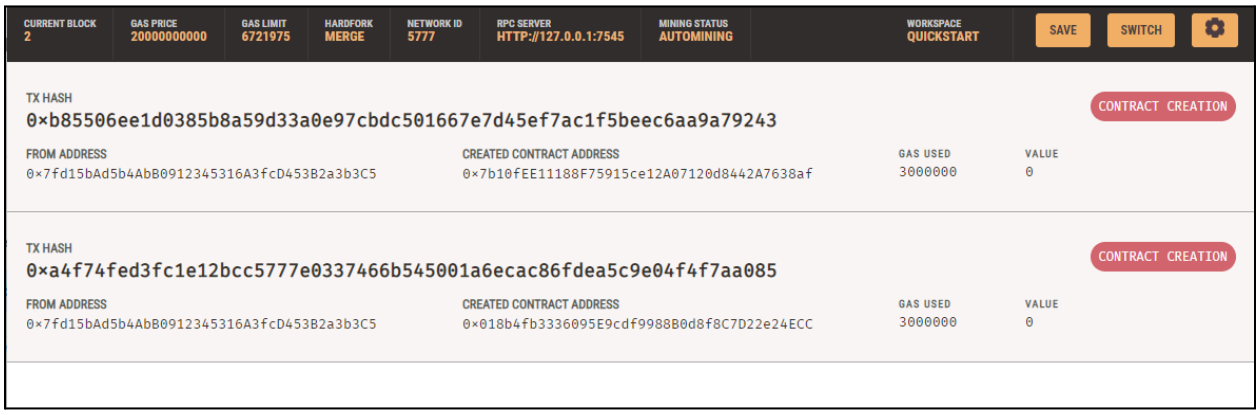
Output -


DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Dev - Ganache Provider
Geth Testnet (1337) network
ACCOUNT
0x7fd...3b3c5 (99.99340007227
0x7fd...3b3c5 (99.99340007227 ETH)
0xa07...ffe24 (100.0 ETH)
0x289...8f426 (100.0 ETH)
0xf3e...934a0 (100.0 ETH)
0x28e...a71e1 (100.0 ETH)
0x0be...b4706 (100.0 ETH)
0x753...751d6 (100.0 ETH)
0xacb...87b36 (100.0 ETH)
0x1a4...2b56d (100.0 ETH)
0x457...cd3dc (100.0 ETH)
DEPLOY & VERIFY
_seller:
0xa07dfedc77ef0cb835a5752c8e81


Add wallet
Private key
Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.
Learn more about how imported keys work
Select type
Private key
Enter your private key string here:
Cancel
Import

ACCOUNT INFORMATION
ACCOUNT ADDRESS
0×A07DFedc77EF0cB835a5752c8e8108E7cA4ffe24
PRIVATE KEY
0×feb942cbaeb4cc850a33f5c87a3e5af51e76d998538bff2fad3f0c643128ed24
Do not use this private key on a public blockchain; use it for development purposes only!
DONE





Conclusion - Solidity smart contract were deployed on remix IDE and transaction were executed successfully on ganache blockchain with the help of metamask wallets.